🧠 OPERATING SYSTEM LAB VIVA – FULL PREP

🔁 SCHEDULING ALGORITHM (General)

Q1. What is CPU scheduling?

Answer:
CPU scheduling is the process of selecting one process from the ready queue to allocate the CPU.

🧠 Smart line:

CPU scheduling maximizes CPU utilization and system throughput.

Q2. Why do we need scheduling algorithms?

Answer:
To:

- Reduce waiting time
- Increase CPU efficiency
- Avoid starvation

Q3. What are scheduling criteria?

Answer:

- CPU utilization
- Throughput
- Turnaround time
- Waiting time
- Response time

🟢 FCFS SCHEDULING (First Come First Serve)

Q1. What is FCFS scheduling?

Answer:
FCFS is a non-preemptive scheduling algorithm where the process that arrives first gets executed first.

Q2. Is FCFS preemptive?

Answer:
No, FCFS is **non-preemptive**.

Q3. Advantages of FCFS?

Answer:

- Simple to implement
- No starvation

Q4. Disadvantages of FCFS?

Answer:

- High waiting time
- Convoy effect

🧠 Smart line:

Short processes may wait behind long processes.

🔵 **ROUND ROBIN SCHEDULING**

Q1. What is Round Robin?

Answer:
Round Robin is a **preemptive scheduling algorithm** where each process gets a fixed time slice (time quantum).

Q2. What is time quantum?

Answer:
Time quantum is the maximum time a process can execute before being preempted.

Q3. What happens if time quantum is too large?

Answer:
It behaves like FCFS.

Q4. Advantages of Round Robin?

Answer:

- Fair scheduling
- No starvation
- Suitable for time-sharing systems

Q5. Disadvantage?

Answer:
Context switching overhead.

🔴 **DEADLOCK**

Q1. What is deadlock?

Answer:
Deadlock is a situation where two or more processes wait indefinitely for resources held by each other.

Q2. What are the 4 necessary conditions of deadlock?

Answer:

1. Mutual Exclusion
2. Hold and Wait

3. No Preemption
4. Circular Wait

🧠 **VERY IMPORTANT** – exam loves this.

Q3. How can deadlock be handled?

Answer:

- Deadlock prevention
- Deadlock avoidance
- Deadlock detection
- Deadlock recovery

🟣 **BANKER'S ALGORITHM**

Q1. What is Banker's Algorithm?

Answer:
Banker's Algorithm is a deadlock avoidance algorithm that checks whether resource allocation leads to a safe state.

Q2. What is a safe state?

Answer:
A state where all processes can complete execution without deadlock.

Q3. What matrices are used in Banker's Algorithm?

Answer:

- Allocation
- Maximum
- Need
- Available

🧠 Formula:

Need = Maximum – Allocation

Q4. Why is it called Banker's Algorithm?

Answer:
Because it works like a bank giving loans only when it is safe.

🟡 **MEMORY ALLOCATION (General)**

Q1. What is memory allocation?

Answer:
Memory allocation is the process of assigning memory blocks to processes.

Q2. Types of memory allocation?

Answer:

- Contiguous
- Non-contiguous

🟠 **FIRST FIT ALGORITHM**

Q1. What is First Fit?

Answer:
First Fit allocates the **first memory block** that is large enough for the process.

Q2. Advantage?

Answer:
Fast allocation.

Q3. Disadvantage?

Answer:
Memory fragmentation.

🟠 **BEST FIT ALGORITHM**

Q1. What is Best Fit?

Answer:
Best Fit allocates the **smallest memory block** that fits the process.

Q2. Advantage?

Answer:
Better memory utilization.

Q3. Disadvantage?

Answer:
Slow because it searches all blocks.

Q4. Which is faster: First Fit or Best Fit?

Answer:
First Fit is faster.

📄 **PAGE REPLACEMENT (General)**

Q1. What is page replacement?

Answer:
Page replacement is used when RAM is full and a new page must be loaded.

Q2. What is a page fault?

Answer:
A page fault occurs when a required page is not present in main memory.

🟤 FIFO PAGE REPLACEMENT

Q1. What is FIFO page replacement?

Answer:
FIFO replaces the **oldest page** in memory.

Q2. Advantage?

Answer:
Easy to implement.

Q3. Disadvantage?

Answer:
Belady's Anomaly.

Q4. What is Belady's Anomaly?

Answer:
Increasing frames can increase page faults.

🧠 Examiner LOVES this question.

💽 DISK SCHEDULING (General)

Q1. What is disk scheduling?

Answer:
Disk scheduling decides the order of servicing disk I/O requests.

Q2. Why is disk scheduling needed?

Answer:
To minimize seek time.

● FCFS DISK SCHEDULING

Q1. What is FCFS disk scheduling?

Answer:
Disk requests are served in the order they arrive.

Q2. Advantage?

Answer:
Simple and fair.

Q3. Disadvantage?

Answer:
High seek time.